AIG-06 · AI IN PROJECT CONTROLS GUIDE

AI for risk identification and quantification

What a model can add to a risk process, and how it manufactures confidence when nobody is watching

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

A method document on AI in project risk work. It separates identification — where machine reading of correspondence, history and the estimate genuinely finds candidate risks a workshop misses — from quantification, where the same tooling can manufacture confidence the inputs do not support. It names what a model cannot see, including the blind spots it inherits from your own register; sets out the six ways a simulation flatters itself; states the decisions that remain human, including the confidence level and contingency release; and works an expected-monetary-value and simulation example in which a single unexamined correlation assumption moves the contingency by more than the smallest risk in the register.

About this document

Document	AIG-06
Series	AI in Project Controls Guide
Version	1.0
Status	draft
Date	2026-08-04
Unresolved placeholders	0

Contents

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Who this is for. Risk managers, cost engineers and planners who maintain quantified registers, project controls managers who set contingency recommendations, and the project directors who approve them.

1. Two failures, and only one of them is new

Risk processes fail in two recognisable ways. The first is old: a register that has become an issues list — things that have already happened, written as risks, with mitigation columns describing work already done. Nothing in it looks forward, and the workshop that produced it drew on the same six people's recent experience.

The second failure is newer and more dangerous: a register that is thin, unexamined and unevidenced, put through a simulation, and returned as a smooth distribution with a contingency figure to the currency unit. Everything about the output signals rigour — the curve, the percentiles, the tornado chart — and none of that rigour is in the inputs. **Simulation transports precision from the arithmetic to the judgement, where it does not belong.**

AI has a genuine contribution to make to the first failure and a strong tendency to worsen the second. Treating identification and quantification as separate problems, with separate controls, is the whole method of this document. The register itself is owned by `BPG-16 — Risk registers that work` and the analysis technique by `BPG-17 — Quantitative schedule risk analysis`; what follows is the AI discipline around them.

2. Identification: what machine reading genuinely adds

Candidate risks from analogous history. Where completed projects have registers and, better, records of which risks actually occurred, a model can propose the risks that materialised on similar work and are absent from this register. The output is a list of candidates for a human to accept, reject or reword — and the rejections matter as much as the acceptances, because they are where the reasoning is recorded.

Risks visible in correspondence. Requests for information (RFIs), non-conformance reports (NCRs), site instructions, minutes and letters carry the early signals of most cost and delay risk: a supplier querying a delivery date, a repeated interface query, an access restriction mentioned in passing. Machine reading covers the whole corpus rather than the sample a human can manage, and that coverage is the strongest argument for AI anywhere in risk work.

Structural gaps. Cross-reading the register against the schedule and estimate surfaces omissions mechanically: long-lead packages with no supply risk, driving-path activities with no risk attached, single-source suppliers with no alternative, consents unrepresented. These are not clever inferences; they are joins nobody has time to do by hand.

Hygiene. De-duplicating near-identical entries, merging two packages' descriptions of the same event, standardising categories and rewriting entries into a disciplined cause–event–effect form are tasks a language model does well and a busy risk manager does last.

3. What identification cannot see

Genuinely novel risk. A model proposes what resembles what it has read. First-of-a-kind work, a new contracting structure, an unprecedented regulatory change or a technology with no delivery record produce nothing to resemble.

Your own blind spots, reproduced. This is the limit that matters most and is named least. A model trained or grounded on your organisation's registers learns your organisation's habits — including the categories you never populate and the risks your culture does not write down. It will return a register that looks complete by your own historical standard, which is precisely the standard in question.

What people choose not to say. Some risks are absent because raising them is uncomfortable: a sponsor's unrealistic date, a partner's capability, a decision already taken. No document contains them, so no model can read them. Surfacing them remains a matter of trust between people, and it is one reason the workshop survives.

Cause. A model can observe that projects with a particular characteristic overran. It cannot establish that the characteristic caused the overrun, and a risk register built on association will attach mitigations to correlates rather than to causes.

4. Quantification: contribution and temptation

The legitimate contributions are real and largely mechanical: proposing three-point ranges from historical spread for review; fitting distributions and reporting the fit; running the simulation with correlations; producing sensitivity and driver rankings; tracking drawdown against the exposure remaining. All of that is computation, and computation is what machines are for.

The temptation is equally specific. Six ways a quantified output flatters itself:

Precision that outruns the inputs. A register scored by judgement in bands returns a P80 to the nearest currency unit. Reporting it that way is a claim about the analysis that the analysis cannot support. Round to the precision the inputs justify and say what that precision is.

Default ranges presented as evidence. Ranges of plus or minus ten per cent applied across a register because that is the template's default produce a distribution built on a habit. Every range should record where it came from: measured spread, supplier quotation, owner judgement, or default — and the proportion that are defaults should be reported alongside the result.

Zero correlation, unexamined. Independence is the default in most tools and is almost never true: productivity risks move together, weather affects several activities, a single supplier fails once and appears in four risks. Independence narrows the distribution and lowers the upper percentiles, which is to say it makes the answer more comfortable and less true. §8 quantifies this on a small register.

A distribution fitted to too few points. Fitting a shape to five historical observations is curve drawing, not estimation. Where evidence is thin, a simple triangular or uniform range honestly labelled as judgement is more defensible than a fitted curve that implies data.

Iteration counts that are not checked for stability. The test is not a number of iterations but whether the reported percentiles are stable when the simulation is re-run. Run it twice; if the P80 moves materially between runs, the result is noise at the precision being reported.

Risks excluded because they cannot be modelled. The unquantifiable risk — a consent refused, a partner withdrawing — drops out of the register because it does not fit the arithmetic, and the total exposure silently excludes the things most likely to matter. Carry them separately and report them alongside the number, explicitly.

To this add the failure mode AI introduces directly: a model asked to **explain** a simulation output will produce a fluent explanation whether or not one exists. A narrative that says the P80 is driven by supply chain and weather is a hypothesis to be checked against the sensitivity output, not a finding.

5. What must not be decided by a model

The confidence level. Whether contingency is set at P50, P80 or elsewhere is a statement of the organisation's risk appetite, made by the accountable authority — not a modelling parameter and not a convention to be inherited from a template.

Contingency release. Drawdown is a management decision against defined criteria, minuted, with the remaining exposure reassessed. A model may report the balance and the exposure; it may not release.

Risk acceptance. Deciding to carry a risk untreated is an accountable judgement about appetite and capacity, not an output of a score.

Whether a risk is closed. Closure means the exposure has genuinely gone — not that the date passed or that nobody has updated the entry. A model may flag stale entries; an owner closes them.

Ownership. Every risk has a named owner who can act. Assignment is a management act; a model proposing an owner from a pattern in the register is proposing, not deciding.

6. Validating a quantified risk output

1. **Confirm the register is a register.** Entries are forward-looking, in cause–event–effect form, with named owners. Issues masquerading as risks are removed to the issues log before anything is quantified.
2. **Trace the inputs.** For each of the largest contributors, confirm probability and impact with the owner, and record the evidence class: measured, quoted, judged or default.

3. **Report the default share.** State what proportion of ranges are template defaults. If it is high, the distribution is a picture of the template.
4. **Test the correlation assumption.** Re-run with a plausible alternative correlation and report the movement. An assumption whose alternative moves the answer materially is a finding, not a setting.
5. **Check stability.** Re-run and confirm the reported percentiles are stable at the precision being reported.
6. **Sanity-check against a deterministic sum.** Compare with the sum of expected monetary values (EMV). The two answer different questions, and the relationship between them should be explicable — see §8.
7. **Check for double counting.** Risk allowances embedded in the estimate, contingency held at package level, and provisions in the forecast frequently overlap. Reconcile explicitly, once, and record it.
8. **List what is excluded.** Every risk left out of the quantification, and why, reported with the result.

7. How this goes wrong

The output is adopted because it is quantified. A number with a distribution behind it beats a number without one in most meetings, regardless of which rests on better inputs. The defence is to present the evidence class of the inputs alongside the result every time.

The model's candidate risks are accepted wholesale. Two hundred proposed risks are pasted into the register to demonstrate thoroughness. The register becomes unusable, owners disengage, and real risks are buried among plausible ones. Accept candidates one at a time, with a reason.

Historical registers are treated as historical outcomes. A register records what people thought might happen, not what did. Learning from registers alone teaches a model the profession's anxieties rather than its experience. Learning from realised risk requires recording, at closeout, which risks actually occurred — a discipline most organisations lack and can start this year.

Correlation is set once and never revisited. The assumption is made during set-up by whoever built the model, is not visible in the output, and survives every subsequent review because nobody knows it is there.

The tornado is read as a cause list. A sensitivity ranking shows which inputs move the answer, and that depends on the ranges assigned as much as on the risks themselves: a risk with a wide default range will outrank a real risk with a tight measured one.

Contingency drifts into a target. Once a P80 figure is in a budget, pressure to reduce it produces quiet changes to input ranges rather than an honest argument about appetite. Version the register, and require that any change to an input range carries the owner's name and reason.

Confidentiality is lost through the register. A quantified register is a map of where a project is weakest and what the organisation believes about its suppliers. It goes into governed tools only.

8. Worked example — EMV, simulation and one unexamined assumption

Illustrative figures. A five-line extract from a quantified cost risk register. Currency USD, all figures at current price basis, impacts stated as most likely cost of occurrence. Not benchmark data.

Risk	Probability	Impact	Expected monetary value
R1 Consent decision later than programmed	0.30	900,000	`0.30 × 900,000 = 270,000`
R2 Rock encountered in bulk excavation	0.45	400,000	`0.45 × 400,000 = 180,000`
R3 Single-source supplier fails to deliver	0.15	1,200,000	`0.15 × 1,200,000 = 180,000`
R4 Installation productivity below assumed rate	0.60	250,000	`0.60 × 250,000 = 150,000`
R5 Currency movement on imported plant	0.35	300,000	`0.35 × 300,000 = 105,000`
Sum of EMV			`270,000 + 180,000 + 180,000 + 150,000 + 105,000 = 885,000`

Step 1 — what the EMV sum is and is not. USD 885,000 is the probability-weighted total. It is not a contingency: no single outcome equals it, and it says nothing about the spread. Its use is as a check on the simulation, not as an answer.

Step 2 — the simulation, as first run. With impacts ranged and risks modelled as independent, the run returns **P50 = 840,000** and **P80 = 1,340,000** (illustrative outputs from the ranges assumed for this example). At the organisation's stated appetite, contingency is recommended at P80.

`Uplift over the EMV sum = 1,340,000 − 885,000 = 455,000`, which is `455,000 ÷ 885,000 = 0.514 = 51.4 %` above the EMV sum

The relationship should be explicable, and here it is. The EMV sum of 885,000 approximates the mean of the simulated total; the P50 of 840,000 sits below it, which is the signature of a right-skewed distribution — the low-probability, high-impact entry (R3) drags the mean up without moving the median. The P80 then sits well above both, because it reaches into that tail. A P80 below the EMV sum is also possible where one rare, very large risk dominates the register: the mean is lifted by an outcome the 80th percentile never reaches. That is a result to investigate and explain, not one to assume is an error.

Step 3 — testing the correlation assumption. R2 and R4 are both ground and productivity related: if the excavation is harder than assumed, installation productivity is likely to suffer too. The analyst re-runs with a correlation of 0.30 between them, leaving every other input untouched. The result:

P80 = 1,455,000, a movement of `1,455,000 − 1,340,000 = 115,000`, or `115,000 ÷ 1,340,000 = 0.086 = 8.6 %` of the original recommendation

Step 4 — read what that means. A single assumption, invisible in the output and made once during set-up, moved the contingency recommendation by USD 115,000 — more than the entire expected monetary value of R5 (USD 105,000), which had a line in the register, an owner and a discussion at the workshop. The correlation had none of those.

Step 5 — what is reported. A contingency recommendation of **USD 1,450,000** — the correlated run, rounded to the nearest 50,000 because the inputs are judgement-based and do not support finer precision — presented with: the confidence level and who set it; the correlation assumption and its effect; the share of impact ranges that are owner-judged rather than measured; and a separate statement of the two risks that could not be sensibly quantified and are therefore excluded from the figure. The recommendation is made by the named risk manager; the release of any part of it is a decision for the project director against the defined criteria.

Assumptions this answer depends on. That probabilities and impacts were confirmed with owners rather than carried forward from the last update; that the impact ranges behind the simulation are as stated and were not template defaults; that the P-values were stable across re-runs; that the 0.30 correlation is a considered judgement about ground conditions and productivity rather than a demonstration figure; and that no part of these five risks is also carried as an allowance inside the base estimate.

9. Checklist — before a quantified risk output informs a decision

1. **Register hygiene done.** Forward-looking entries, cause–event–effect wording, named owners, issues removed to the issues log.
2. **Top contributors confirmed with owners** in the current period, not inherited.
3. **Evidence class recorded** for every impact range: measured, quoted, judged or default — and the default share reported.
4. **Correlation stated and tested**, with the movement from a plausible alternative reported.
5. **Stability confirmed** by re-running and comparing the reported percentiles.
6. **EMV cross-check done**, and the relationship between the EMV sum and the chosen percentile explained.
7. **Double counting reconciled** across base estimate allowances, package contingency and the forecast.
8. **Exclusions listed** — every risk left out of the quantification and why.
9. **Precision honest** — rounded to what the inputs support, with the confidence level and its owner stated.
10. **Decisions attributed.** The confidence level, the recommendation and any release each carry a name.

Related

- `AIG-04 — AI-assisted cost forecasting` — where contingency and forecast meet, and the double-counting check at §6.7.
- `AIG-05 — AI in scheduling — and what must not be automated` — the schedule ranges that feed quantitative schedule risk analysis.
- `BPG-16 — Risk registers that work` — the register discipline this document assumes.
- `BPG-17 — Quantitative schedule risk analysis` — the analysis technique, in method detail.

- `TPL-10 — Risk register` — the instrument, including the evidence-class and correlation fields §6 relies on.

Sources and standards

- **ISO 31000:2018**, Risk management — Guidelines. In our own words: it describes risk management as an integrated, structured activity built on defined context, explicit criteria, and identification, analysis, evaluation and treatment steps, with information quality and human judgement treated as part of the process rather than as inputs to a calculation. Its insistence that criteria are set before analysis is the principle behind §5.
- **PCI Body of Knowledge, Domain 12** — Risk management for project controls (Institute manuscript, 2026). Expected monetary value, quantification and contingency setting as taught by the Institute.
- **PCI Body of Knowledge, Domain 13** — AI for project controls and project management (Institute manuscript, 2026), Knowledge Area 13.5.9, the risk workflow and its warning about contingency set by an unexamined algorithm.

Status and version

Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.